

Tianyu Shen

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Education

Fudan University

Sep. 2024 – Jun. 2028

B.S. in Computer Science

GPA: 3.96 / 4.00 Rank: 2 / 101

University of North Carolina at Chapel Hill

Aug. 2026 – Jan. 2027

Exchange Student

Research Experience

Fine-grained Referring Audio Separation

Apr. 2026 – Present

RAIR Lab; advised by Prof. Jason Ren, UNC Chapel Hill

- Investigating **fine-grained referring audio separation**, where natural-language queries identify specific sound sources through visual and contextual cues.
- Designing a large-scale benchmark spanning **music, speech, and general sounds**, primarily from real-world audiovisual data.
- Building data pipelines for sound-source selection, reference-audio construction, synthetic mixture generation, and fine-grained query generation.
- Evaluating **SAM-Audio** across different referring conditions and analyzing failure modes of existing audio separation models.
- Investigating methods to improve fine-grained referring performance based on our benchmark; *manuscript in preparation*.

Decentralized Social Network Analysis

Oct. 2025 – Jan. 2026

Big Data and Networking Group; advised by Prof. Yang Chen, Fudan University

- Analyzed large-scale interaction patterns in decentralized social networks (e.g., Mastodon) using the FediLive framework.
- Developed network-based models for multi-layer interactions across instances and contributed to a collaborative research paper on decentralized governance.

AI Agent for Degree Thesis Submission & Consultation

Apr. 2025 – May 2025

Information Office, Fudan University

- Designed a dual-agent LLM system for student- and faculty-facing degree thesis consultation.
- Built a Retrieval-Augmented Generation pipeline and deployed the system for real university administrative scenarios.

Awards

Fudan University Undergraduate Scholarship, Third Prize

Oct. 2025

Fudan University AI Application Development Competition, Third Prize

May 2025

Fudan University Freshman Scholarship

Sep. 2024

Technical Skills

Machine Learning: PyTorch, deep learning, multimodal learning, audio processing, source separation.

Tools: Hugging Face, Git, Linux, FFmpeg, Jupyter.

Programming: Python, C/C++, MATLAB, Verilog, L^AT_EX.

Languages: Mandarin (Native), English (Fluent; TOEFL 112: Speaking 28, Writing 28).